

Introduction

Document structure extraction is an important task in NLP.

It enables downstream applications like RAG and information retrieval.

Background

Existing methods can be categorized into rule-based and learning-based.

Rule-Based Methods

Rule-based methods use font size, bold, and numbering patterns.

They work well on well-formatted documents but fail on plain text.

Learning-Based Methods

Learning-based methods use neural networks to predict heading boundaries.

They require large amounts of labeled training data.

Our Approach

We propose a hybrid method that uses LLM for semantic understanding.

Skeleton Compression

The document is compressed to 10-30 percent of its original size.

This reduces LLM API costs significantly.

Cursor Routing

The LLM annotates section boundaries on the compressed skeleton.

It outputs `block_id`, `level`, and `title` for each heading.

Deterministic Assembly

A deterministic algorithm reconstructs the full document tree.

This guarantees zero content loss.

Evaluation

We test on 12 academic papers and 9 benchmark documents.

Our method achieves 2.6x higher F1 than pure LLM E2E.

Conclusion

Control-dataflow decoupling is the key to reliable document parsing.